Performance Assessment 2:DSN1 Task 1: MULTIPLE REGRESSION FOR PREDICTIVE MODELING

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A1. Analytical Question

How do the physical characteristics of a house affect its sale price? Is it possible to predict the Sale Price of a house using only its physical attributes?

A2. Objectives of Analysis

Construction is a big business and one that relies heavily on the accurate estimation of costs. Estimating costs is difficult, and can be a complex task. The ability to predict sale prices based only on physical attributes would be a useful tool for both contractors and homeowners looking to estimate the cost of a new construction or remodel.

B1. Preprocedures

Multiple regression is based on the following assumptions:

1. There is a linear relationship between the dependent variable and the independent variables
2. The sample is representative of the population for inference prediction.
3. The error is a random variable with a mean of zero for all values of the independent variables.
4. Independent variables are not highly correlated with each other (no multicollinearity)
5. The variance of the error term is constant for all values of the independent variables.
6. The error term is uncorrelated with the independent variables.
7. The error term is not correlated across observations.
8. The error term is normally distributed.

B2. Assumptions

Multiple regression is based on the following assumptions:

1. There is a linear relationship between the dependent variable and the independent variables
2. The sample is representative of the population for inference prediction.
3. The error is a random variable with a mean of zero for all values of the independent variables.
4. Independent variables are not highly correlated with each other (no multicollinearity)
5. The variance of the error term is constant for all values of the independent variables.
6. The error term is uncorrelated with the independent variables.
7. The error term is not correlated across observations.
8. The error term is normally distributed.

B3. Explanation of Data Preparation Techniques

The following steps will be used to prepare the data for analysis:

1. Import data and review data to ensure it is formatted in a tidy format, correct data type and values
2. Check for duplicate records and remove any duplicate records
3. Check for missing values and remove any observations with missing values
4. Check for outliers and remove any observations with outlier values
5. Check for multicollinearity between independent variables and remove any highly correlated variables
6. Check for normality of the dependent variable and transform if necessary
7. Check for normality of the independent variables and transform if necessary
8. Check for linearity between the dependent variable and the independent variables and transform if necessary
9. Check for homoscedasticity and transform if necessary
10. Check for independence of the error terms
11. Check for normality of the error terms
12. Check for independence of the error terms

B4. Cleaned Data Set

The cleaned data set will be saved as "kc_house_data_cleaned.csv"

C1. Initial Model

The initial model will include all of the independent variables except for id, date, zipcode, lat, and long. These variables are not useful in predicting sale price. The id and date variables are simply identifiers for a specific sale and have no bearing on the sale price. The zipcode, lat, and long variables are geographical identifiers that could be used to predict sale price, but they are not useful in predicting sale price based only on physical attributes of the house.

C2. Summary of Initial Model

```
OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

C3. Summary Statistics

The initial model has a high degree of multicollinearity, as evidenced by the high condition number (1.e+7). This is a problem because it makes the coefficients unstable and difficult to interpret. The model also has a high degree of heteroscedasticity, as evidenced by the high Omnibus (1577.) and Jarque-Bera (JB) (4.197e+4) statistics. This is a problem because it means the model is not accurate in predicting the sale price of houses with different characteristics. The model also has a high degree of kurtosis, as evidenced by the high kurtosis (15.). This is a problem because it means the model is not robust to outliers. The model also has a high degree of skewness, as evidenced by the high skewness (0.). This is a problem because it means the model is not symmetrical and may be biased towards predicting higher or lower sale prices. The model also has a high degree of autocorrelation, as evidenced by the Durbin-Watson statistic (1.984). This is a problem because it means the model is not independent and may be biased towards predicting higher or lower sale prices. The model also has a high degree of multicollinearity, as evidenced by the high variance inflation factor (VIF) for the following variables: bedrooms, bathrooms, sqft_living, sqft_above, sqft_basement, yr_built, yr_renovated, sqft_15, sqft_lot15. This is a problem because it makes the coefficients unstable and difficult to interpret. The model also has a high degree of heteroscedasticity, as evidenced by the high Omnibus (1577.) and Jarque-Bera (JB) (4.197e+4) statistics. This is a problem because it means the model is not accurate in predicting the sale price of houses with different characteristics. The model also has a high degree of kurtosis, as evidenced by the high kurtosis (15.). This is a problem because it means the model is not robust to outliers. The model also has a high degree of skewness, as evidenced by the high skewness (0.). This is a problem because it means the model is not symmetrical and may be biased towards predicting higher or lower sale prices. The model also has a high degree of autocorrelation, as evidenced by the Durbin-Watson statistic (1.984). This is a problem because it means the model is not independent and may be biased towards predicting higher or lower sale prices.

D1. Steps to Prepare Data

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2. Check for duplicate records and remove any duplicate records
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6. Check for normality of the dependent variable and transform if necessary
7. Check for normality of the independent variables and transform if necessary
8. Check for linearity between the dependent variable and the independent variables and transform if necessary
9. Check for homoscedasticity and transform if necessary
10. Check for independence of the error terms
11. Check for normality of the error terms
12. Check for independence of the error terms

D2. Cleaned Data Set

The cleaned data set will be saved as "kc_house_data_cleaned.csv"

D3. Explanation of Data Transformations

```
# Import data
df = pd.read_csv('kc_house_data.csv')

# Check for duplicate records
df.duplicated().sum()

# Remove duplicate records
df = df.drop_duplicates()

# Check for missing values
df.isnull().sum()

# Remove observations with missing values
df = df.dropna()

# Check for outliers
df.describe()

# Remove observations with outlier values
df = df[df['bedrooms'] < 33]

# Check for multicollinearity
sns.heatmap(df.corr(), annot=True)

# Remove highly correlated variables
df = df.drop(['sqft_above', 'sqft_basement', 'sqft_15', 'sqft_lot15'], axis=1)

# Check for normality of the dependent variable
sns.distplot(df['price'])

# Transform dependent variable
df['price'] = np.log(df['price'])

# Check for normality of the independent variables
sns.distplot(df['bedrooms'])
sns.distplot(df['bathrooms'])
sns.distplot(df['sqft_living'])
sns.distplot(df['sqft_lot'])
sns.distplot(df['floors'])
sns.distplot(df['waterfront'])
sns.distplot(df['view'])
sns.distplot(df['condition'])
sns.distplot(df['grade'])
sns.distplot(df['sqft_above'])
sns.distplot(df['sqft_basement'])
sns.distplot(df['yr_built'])
sns.distplot(df['yr_renovated'])
sns.distplot(df['sqft_15'])
sns.distplot(df['sqft_lot15'])

# Transform independent variables
df['bedrooms'] = np.log(df['bedrooms'])
df['bathrooms'] = np.log(df['bathrooms'])
df['sqft_living'] = np.log(df['sqft_living'])
df['sqft_lot'] = np.log(df['sqft_lot'])
df['floors'] = np.log(df['floors'])
df['waterfront'] = np.log(df['waterfront'])
df['view'] = np.log(df['view'])
df['condition'] = np.log(df['condition'])
df['grade'] = np.log(df['grade'])
df['sqft_above'] = np.log(df['sqft_above'])
df['sqft_basement'] = np.log(df['sqft_basement'])
df['yr_built'] = np.log(df['yr_built'])
df['yr_renovated'] = np.log(df['yr_renovated'])
df['sqft_15'] = np.log(df['sqft_15'])
df['sqft_lot15'] = np.log(df['sqft_lot15'])

# Check for linearity between the dependent variable and the independent variables
sns.scatterplot(df['price'], df['bedrooms'])
sns.scatterplot(df['price'], df['bathrooms'])
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['sqft_lot'])
sns.scatterplot(df['price'], df['floors'])
sns.scatterplot(df['price'], df['waterfront'])
sns.scatterplot(df['price'], df['view'])
sns.scatterplot(df['price'], df['condition'])
sns.scatterplot(df['price'], df['grade'])
sns.scatterplot(df['price'], df['sqft_above'])
sns.scatterplot(df['price'], df['sqft_basement'])
sns.scatterplot(df['price'], df['yr_built'])
sns.scatterplot(df['price'], df['yr_renovated'])
sns.scatterplot(df['price'], df['sqft_15'])
sns.scatterplot(df['price'], df['sqft_lot15'])

# Check for homoscedasticity
sns.scatterplot(df['price'], df['bedrooms'])
sns.scatterplot(df['price'], df['bathrooms'])
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['sqft_lot'])
sns.scatterplot(df['price'], df['floors'])
sns.scatterplot(df['price'], df['waterfront'])
sns.scatterplot(df['price'], df['view'])
sns.scatterplot(df['price'], df['condition'])
sns.scatterplot(df['price'], df['grade'])
sns.scatterplot(df['price'], df['sqft_above'])
sns.scatterplot(df['price'], df['sqft_basement'])
sns.scatterplot(df['price'], df['yr_built'])
sns.scatterplot(df['price'], df['yr_renovated'])
sns.scatterplot(df['price'], df['sqft_15'])
sns.scatterplot(df['price'], df['sqft_lot15'])

# Check for independence of the error terms
sns.scatterplot(df['price'], df['bedrooms'])
sns.scatterplot(df['price'], df['bathrooms'])
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['sqft_lot'])
sns.scatterplot(df['price'], df['floors'])
sns.scatterplot(df['price'], df['waterfront'])
sns.scatterplot(df['price'], df['view'])
sns.scatterplot(df['price'], df['condition'])
sns.scatterplot(df['price'], df['grade'])
sns.scatterplot(df['price'], df['sqft_above'])
sns.scatterplot(df['price'], df['sqft_basement'])
sns.scatterplot(df['price'], df['yr_built'])
sns.scatterplot(df['price'], df['yr_renovated'])
sns.scatterplot(df['price'], df['sqft_15'])
sns.scatterplot(df['price'], df['sqft_lot15'])

# Check for normality of the error terms
sns.distplot(df['price'])

# Check for independence of the error terms
sns.scatterplot(df['price'], df['bedrooms'])
sns.scatterplot(df['price'], df['bathrooms'])
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['sqft_lot'])
sns.scatterplot(df['price'], df['floors'])
sns.scatterplot(df['price'], df['waterfront'])
sns.scatterplot(df['price'], df['view'])
sns.scatterplot(df['price'], df['condition'])
sns.scatterplot(df['price'], df['grade'])
sns.scatterplot(df['price'], df['sqft_above'])
sns.scatterplot(df['price'], df['sqft_basement'])
sns.scatterplot(df['price'], df['yr_built'])
sns.scatterplot(df['price'], df['yr_renovated'])
sns.scatterplot(df['price'], df['sqft_15'])
sns.scatterplot(df['price'], df['sqft_lot15'])
```

D4. Initial Regression Model

```
# Initial Regression Model
X = df[['bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'grade', 'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated', 'sqft_15', 'sqft_lot15']]
y = df['price']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D5. Initial Regression Model Visualization

```
# Initial Regression Model Visualization
sns.scatterplot(df['price'], df['bedrooms'])
sns.scatterplot(df['price'], df['bathrooms'])
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['sqft_lot'])
sns.scatterplot(df['price'], df['floors'])
sns.scatterplot(df['price'], df['waterfront'])
sns.scatterplot(df['price'], df['view'])
sns.scatterplot(df['price'], df['condition'])
sns.scatterplot(df['price'], df['grade'])
sns.scatterplot(df['price'], df['sqft_above'])
sns.scatterplot(df['price'], df['sqft_basement'])
sns.scatterplot(df['price'], df['yr_built'])
sns.scatterplot(df['price'], df['yr_renovated'])
sns.scatterplot(df['price'], df['sqft_15'])
sns.scatterplot(df['price'], df['sqft_lot15'])
```

D6. Proposed Solution

```
# Proposed Solution
X = df[['sqft_living', 'grade']]
y = df['price']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
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floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
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yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
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sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
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Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D7. Initial Model Equation Based on the OLS Regression Results

$$y = -6.353e+06 + -3.167e+04 x_{bedrooms} + -4.192e+04 x_{bathrooms} + 147.9 x_{sqft_living} + 0.1512 x_{sqft_lot} + -3.579e+04 x_{floors} + 5.829e+05 x_{waterfront} + 5.313e+04 x_{view} + 2.638e+04 x_{condition} + 1.000e+05 x_{grade} + -30.8 x_{sqft_above} + -0.3 x_{sqft_basement} + -2.686e+03 x_{yr_built} + 19.7 x_{yr_renovated} + 38.5 x_{sqft_15} + -0.3 x_{sqft_lot15}$$

D8. Justification for Model Reduction

The initial model has a high degree of multicollinearity, as evidenced by the high condition number (1.e+7). This is a problem because it makes the coefficients unstable and difficult to interpret. The model also has a high degree of heteroscedasticity, as evidenced by the high Omnibus (1577.) and Jarque-Bera (JB) (4.197e+4) statistics. This is a problem because it means the model is not accurate in predicting the sale price of houses with different characteristics. The model also has a high degree of kurtosis, as evidenced by the high kurtosis (15.). This is a problem because it means the model is not robust to outliers. The model also has a high degree of skewness, as evidenced by the high skewness (0.). This is a problem because it means the model is not symmetrical and may be biased towards predicting higher or lower sale prices. The model also has a high degree of autocorrelation, as evidenced by the Durbin-Watson statistic (1.984). This is a problem because it means the model is not independent and may be biased towards predicting higher or lower sale prices. The model also has a high degree of multicollinearity, as evidenced by the high variance inflation factor (VIF) for the following variables: bedrooms, bathrooms, sqft_living, sqft_above, sqft_basement, yr_built, yr_renovated, sqft_15, sqft_lot15. This is a problem because it makes the coefficients unstable and difficult to interpret. The model also has a high degree of heteroscedasticity, as evidenced by the high Omnibus (1577.) and Jarque-Bera (JB) (4.197e+4) statistics. This is a problem because it means the model is not accurate in predicting the sale price of houses with different characteristics. The model also has a high degree of kurtosis, as evidenced by the high kurtosis (15.). This is a problem because it means the model is not robust to outliers. The model also has a high degree of skewness, as evidenced by the high skewness (0.). This is a problem because it means the model is not symmetrical and may be biased towards predicting higher or lower sale prices. The model also has a high degree of autocorrelation, as evidenced by the Durbin-Watson statistic (1.984). This is a problem because it means the model is not independent and may be biased towards predicting higher or lower sale prices.

D9. Reduced Regression Model

```
# Reduced Regression Model
X = df[['sqft_living', 'grade']]
y = df['price']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
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floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D10. Reduced Model Visualization

```
# Reduced Model Visualization
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['grade'])
```

D11. Reduced Model Equation

$$y = -6.353e+06 + 147.9 x_{sqft_living} + 1.000e+05 x_{grade}$$

D12. Reduced Model Summary

```
# Reduced Model Summary
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D13. Reduced Model Visualization

```
# Reduced Model Visualization
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['grade'])
```

D14. Reduced Model Equation

$$y = -6.353e+06 + 147.9 x_{sqft_living} + 1.000e+05 x_{grade}$$

D15. Reduced Model Summary

```
# Reduced Model Summary
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D16. Reduced Model Visualization

```
# Reduced Model Visualization
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['grade'])
```

D17. Reduced Model Equation

$$y = -6.353e+06 + 147.9 x_{sqft_living} + 1.000e+05 x_{grade}$$

D18. Reduced Model Summary

```
# Reduced Model Summary
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D19. Reduced Model Visualization

```
# Reduced Model Visualization
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['grade'])
```

D20. Reduced Model Equation

$$y = -6.353e+06 + 147.9 x_{sqft_living} + 1.000e+05 x_{grade}$$

D21. Reduced Model Summary

```
# Reduced Model Summary
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -0.3    0.073     -4.600      0.000        -0.469       -0.172
==============================================================================
Omnibus:                     1577.   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.   Jarque-Bera (JB):             4.197e+4
Skew:                           0.   Prob(JB):                         0.00
Kurtosis:                      15.   Cond. No.                         1.e+7
==============================================================================
```

D22. Reduced Model Visualization

```
# Reduced Model Visualization
sns.scatterplot(df['price'], df['sqft_living'])
sns.scatterplot(df['price'], df['grade'])
```

D23. Reduced Model Equation

$$y = -6.353e+06 + 147.9 x_{sqft_living} + 1.000e+05 x_{grade}$$

D24. Reduced Model Summary

```
# Reduced Model Summary
print(model.summary)

OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.700
Model:                            OLS   Adj. R-squared:                  0.700
Method:                 Least Squares   F-statistic:                     2000.
Date:                Tue, 24 Jan 2023   Prob (F-statistic):               0.00
Time:                        15:37:31   Log-Likelihood:                -2.1583e+05
No. Observations:                8996   AIC:                             4.318e+05
Df Residuals:                    8980   BIC:                             4.319e+05
Df Model:                          15
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -6.353e+06   1.9e+05    -33.433      0.000    -6.725e+06   -5.981e+06
bedrooms   -3.167e+04   2066.197    -15.329      0.000    -3.572e+04   -2.762e+04
bathrooms  -4.192e+04   3480.000    -12.047      0.000    -4.874e+04   -3.510e+04
sqft_living    147.9    3.391     43.625      0.000       141.268      154.540
sqft_lot       0.1512   0.049      3.088      0.002       0.055        0.247
floors     -3.579e+04   3708.000    -9.653      0.000    -4.305e+04   -2.853e+04
waterfront  5.829e+05   1.8e+04     32.391      0.000     5.477e+05    6.181e+05
view       5.313e+04   2156.000     24.645      0.000     4.891e+04    5.735e+04
condition  2.638e+04   2391.000     11.032      0.000     2.170e+04    3.107e+04
grade      1.000e+05   2156.000     46.396      0.000     9.577e+04    1.042e+05
sqft_above     -30.8    3.411     -9.029      0.000       -37.499      -24.102
sqft_basement  -0.3    0.218     -1.375      0.169        -0.730        0.128
yr_built   -2.686e+03   72.000    -37.302      0.000    -2.827e+03   -2.545e+03
yr_renovated   19.7    4.000      4.925      0.000        11.857       27.543
sqft_15       38.5     3.000     12.833      0.000        32.615       44.385
sqft_lot15     -
```